



**ENT-164** · Intro to Making

**Tufts**  
UNIVERSITY

Derby Entrepreneurship  
Center at Tufts

CLASS 3 · WEEK 3 · FALL 2026

# From 2D to 3D

## Laser Cutting

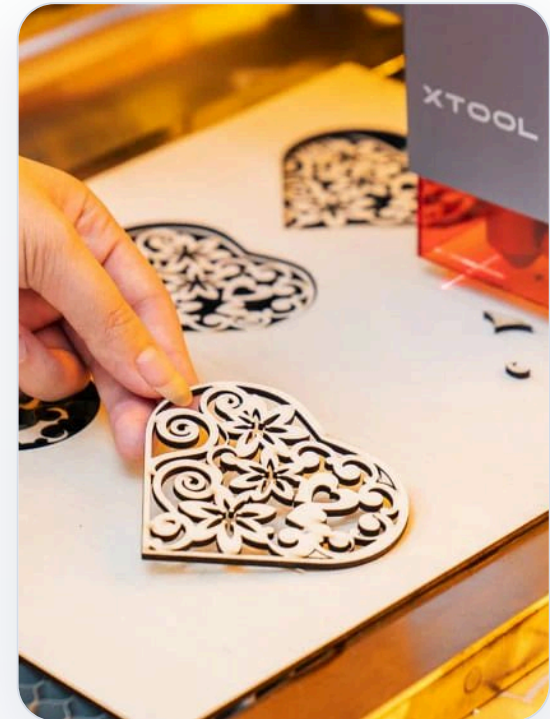
Turn your robot sketches and measurements into a laser-ready 2D design. Measure the real parts, model in Onshape, and cut at Nolop.

Thu, Sep 24

JCC 301 / Nolop

Onshape + Inkscape

Individual laser-cut project



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Tufts University · Gordon Institute

# Seven weeks from sketch to race day

Today is Week 3. From Physical to Digital: 2D. Here's how the robot build comes together.

## Week 2

### Ideation and Prototyping

Sketch concept variations; build a foam core mockup.

## Week 3

### From Physical to Digital: 2D

Create a laser-ready 2D design; cut structural parts.

## Week 4

### From Physical to Digital: 3D

Model 3D parts; 3D print functional parts.

## Week 5

### Basic Electronics and Microcontrollers

Wire motors, sensors, and controls; program with MicroPython + AI.

## Week 6

### AI Coding Agents

Connect to a wireless network; build a remote control interface.

## Week 7

### Final Robot Assembly and Testing

Final integration and test runs.

## Week 8

### Robo Challenge and Final Project Kickoff

Demonstrate the robot with mechanical systems and motors mounted.

## TODAY

**Week 3 · Laser Cutting.** Go from a foam core prototype to a laser-cut assembly: measure the real parts, draw them in Onshape, and cut.

# From mockup to laser-cut assembly

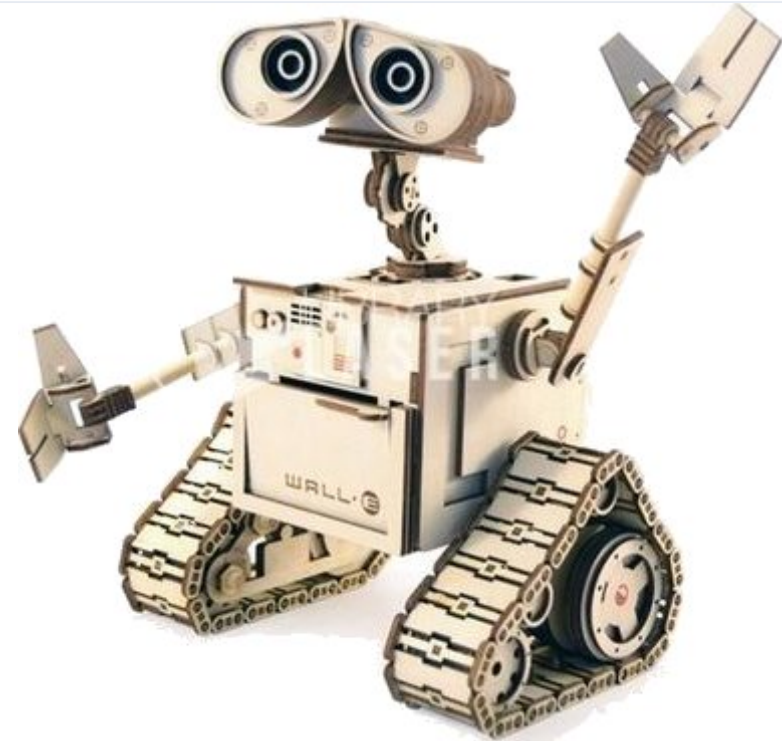
Your foam core prototype proved the idea. Now we design the real parts.

## TODAY'S PLAN

- Read your sketches as engineering drawings
- Pull numbers from datasheets
- Measure the real parts with calipers
- Draw the parts in Onshape
- Turn a sketch into a laser-ready DXF

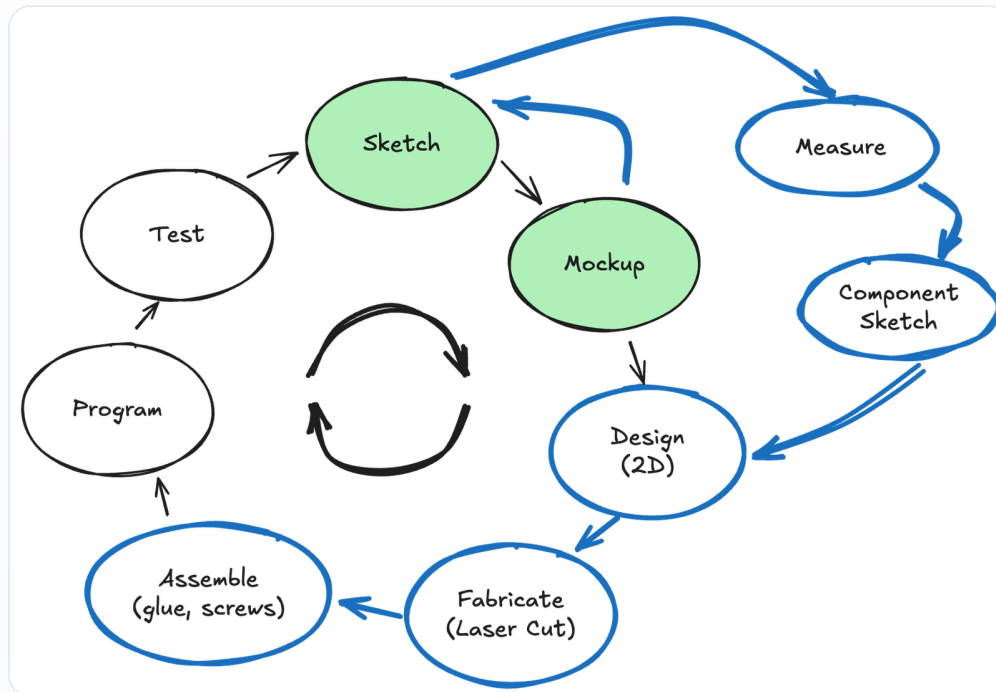
## WHERE WE ARE IN THE BUILD

Sketch and mockup are behind you. Today the design becomes a real, cut part, then we fabricate, assemble, and get on to electronics.



## Sketch, mockup, design, fabricate

2D design sits between the mockup and the finished part, and the cycle repeats.



## Measure before you draw

Every hole and every slot starts with a real number from a real part. Never a guess.

## Design for the sheet

Laser cutting works on flat stock, so think in 2D: flat panels that slot, bolt, or glue into a 3D object.

## Fabricate, then check

Cut a test part in cardboard before you commit a sheet of basswood to the design.

# Sign up for Onshape

Onshape is a CAD program that runs in your browser. Nothing to install, and your work is always saved.

## DO THIS NOW

Sign up with your **Tufts account** to get the free education plan.

[onshape.com/en/education/sign-up](https://onshape.com/en/education/sign-up)

## Why Onshape?

It runs in the browser on any laptop, keeps a version history of every change, and exports a DXF that the laser cutter can read.



Use your Tufts account. The education plan is free.

# The top view is your **base plate**

Looking straight down: the footprint of the robot, the wheel positions, and the mounting holes for the motors.

## Overall size

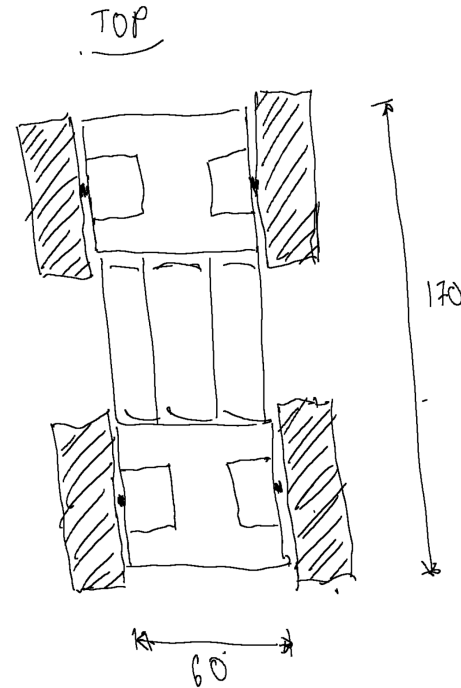
170 mm long and 60 mm wide set the sheet footprint of the whole assembly.

## Wheel cutouts

The hatched panels at each corner are the wheels, bolted through the base plate.

## Motor mounts

The two small squares are the motor housings. Their spacing comes from the datasheet.



# The side view sets the heights

Looking from the side: how tall the body is, where the electronics sit, and how the wheel clears the deck.

## Body height

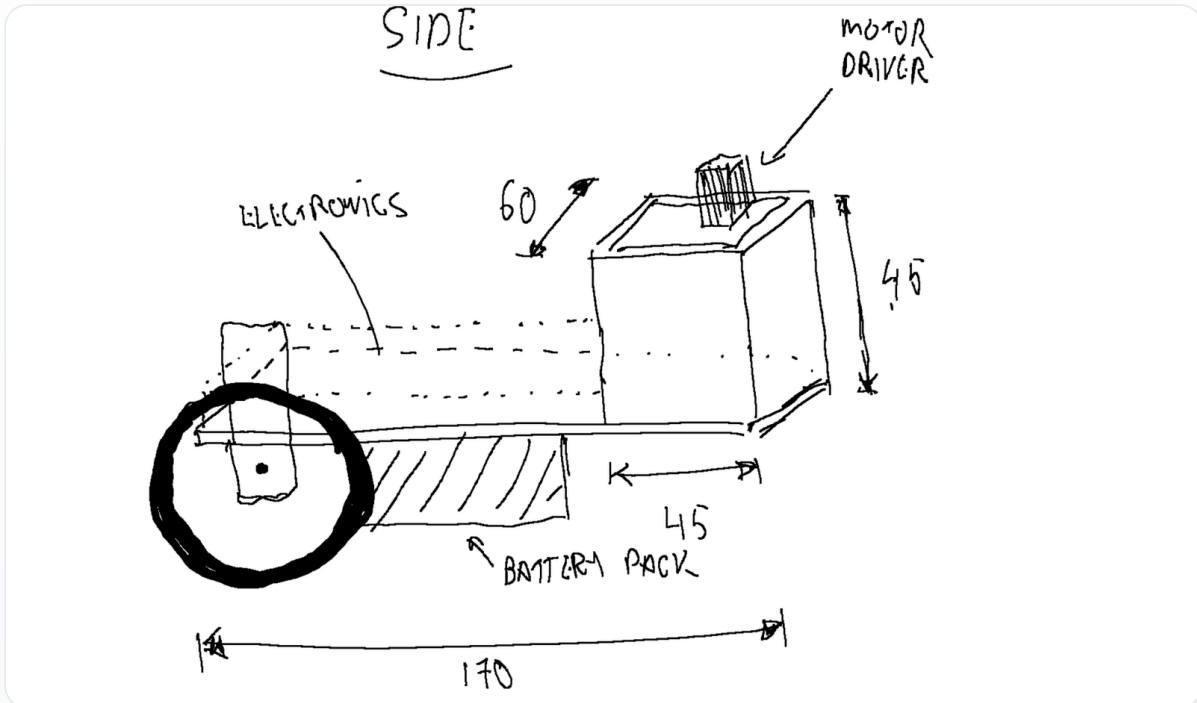
45 mm of clearance under the deck for the electronics and the motor driver.

## Battery pack

45 mm long and 60 mm wide. It decides where the electronics panel sits.

## Wheel diameter

The wheel is the lowest point, so it sets how the chassis meets the ground.



# The axle is a **measured part**

Small parts need the tightest numbers. The axle is where a loose hole turns into a wobbly wheel.

## Axle length

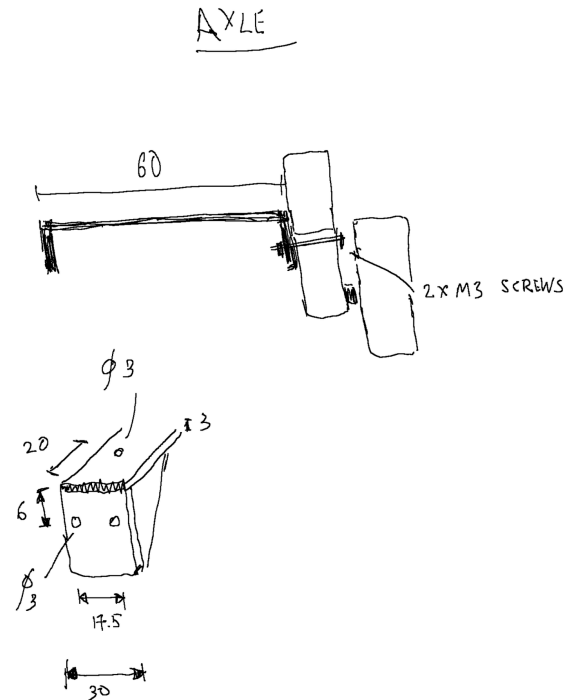
60 mm from the motor to the far wheel, held by 2 × M3 screws.

## Hole sizes

A Ø3 shaft hole and Ø3 mounting holes at 17.5 mm spacing. Half of a 35 mm motor centre.

## Material thickness

The class cuts 3 mm basswood sheets. Every slot you draw has to match the material it holds.



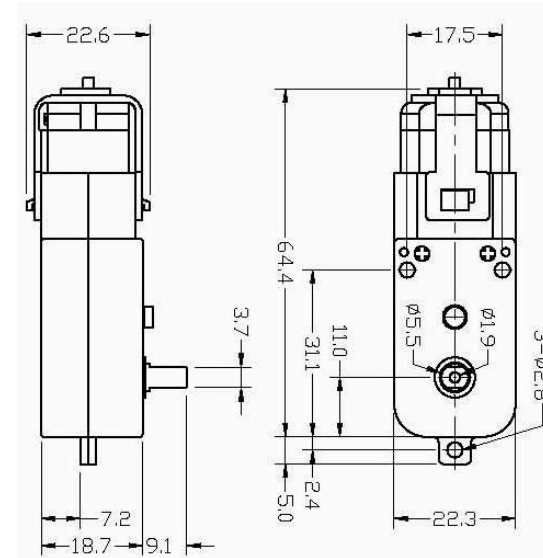
# Leverage datasheets

Every part you bought came with a drawing. Use it. The manufacturer already measured the part for you.

## THE FOUR NUMBERS YOU NEED

- **Mounting hole distance.** 17.5 Mm and 22.3 mm on the TT motor
- **Mounting hole diameter.**  $\varnothing 2.4$  /  $\varnothing 3$  Mm, depending on the screw
- **Motor housing height.** 64.4 Mm overall, 36.8 mm to the shaft
- **Motor housing depth.** 22.6 Mm, plus the 9.1 mm shaft

Check the drawing for **hole spacing** before you draw the mount. That single number decides whether the motor fits.



A TT motor datasheet. All the numbers for the mount are already here.

# Take your own measurements

When there's no datasheet, or the part is already in your hand, the caliper is the answer.

## Outer jaws (1)

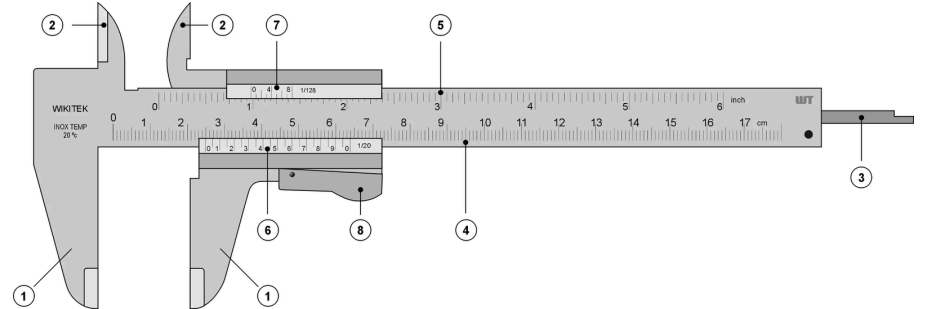
Measure the outside of a shaft, a wheel hub, or sheet thickness.

## Inner jaws (2)

Measure the inside of a hole or a slot. The two small jaws at the top.

## Depth rod (3)

Reach into a bore to find how deep a part really sits. Handy for the motor boss.



## TIP

Measure twice, and write the number on the sketch before you open Onshape.

# Bolts, washers, and nuts

Flat parts become a 3D object when you bolt them together, so your drawing needs the hole, not only the shape.

## The bolt sizes the hole

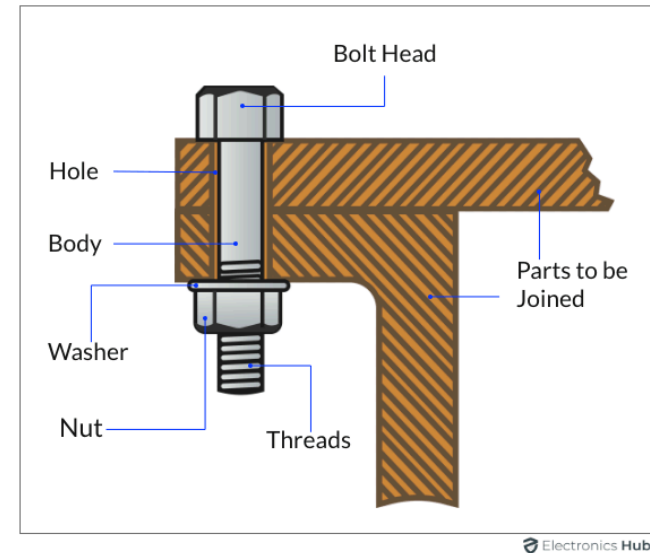
An M3 bolt needs a clearance hole around 3.2 mm in the parts it passes through. The head and the nut do the clamping.

## The washer spreads the load

Under the nut and the head, it stops the bolt from crushing soft basswood or cracking acrylic.

## The nut locks it

Threads only bite where there is a matching nut, so plan for access on both sides of the joint.



A bolted joint: head, hole, washer, nut, and threads.

# Screws that make **their own hole**

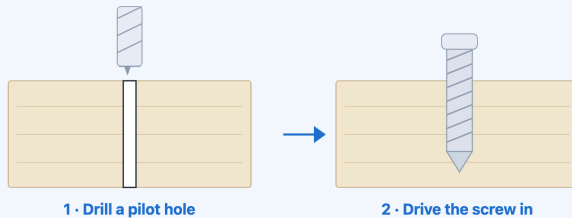
Two screw types you'll use on wood and acrylic, and the pilot hole each one needs.

01

## Self-tapping

Needs a **pre-drilled pilot hole**. The screw cuts its own threads as it goes in. A pilot hole slightly smaller than the screw keeps the wood from splitting.

Use it when the screw has to grip the material directly, with no nut on the far side.

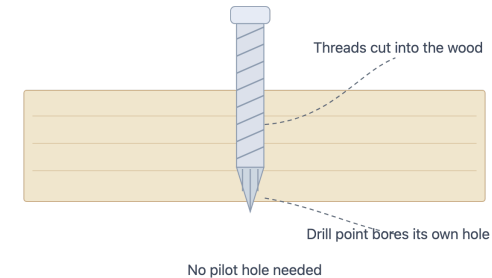


02

## Self-drilling

Has a **drill point on the tip**. It drills its own hole and taps its own threads in one pass. No pilot hole needed.

Fast for thin sheet and soft material, but it can split thin basswood or crack acrylic.



### ON THE LASER

**You can't drill a hole with a laser.** Laser-cut a pilot hole slightly undersized and let the screw cut its own thread, or cut the clearance hole and use a bolt, washer, and nut.

# Draw your first laser-ready shapes

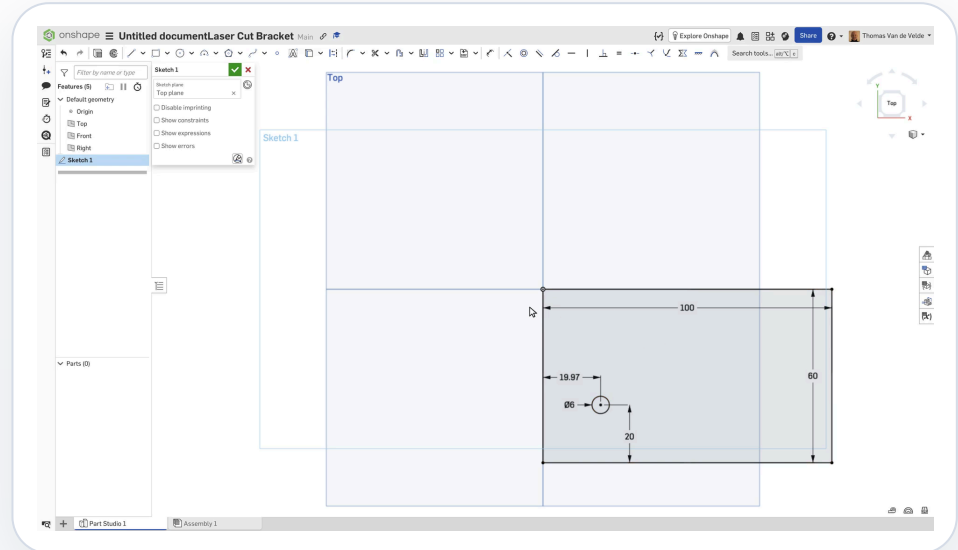
Follow along in your own Onshape document. Five shapes cover almost every part you'll cut.

## IN THIS DEMO

- **A workspace overview.** Documents, tabs, and the feature list
- **A basic rectangle.** Start a sketch and draw the base plate
- **Updating dimensions.** Drive every size from a typed number
- **Mirroring.** One wheel mount becomes two
- **Combining shapes and the Trim tool,** slots and cutouts
- **Creating joints,** check that the parts assemble

## From sketch to laser file

Right-click the sketch in the feature list and choose **Export as DXF/DWG**. That file goes to the laser cutter.

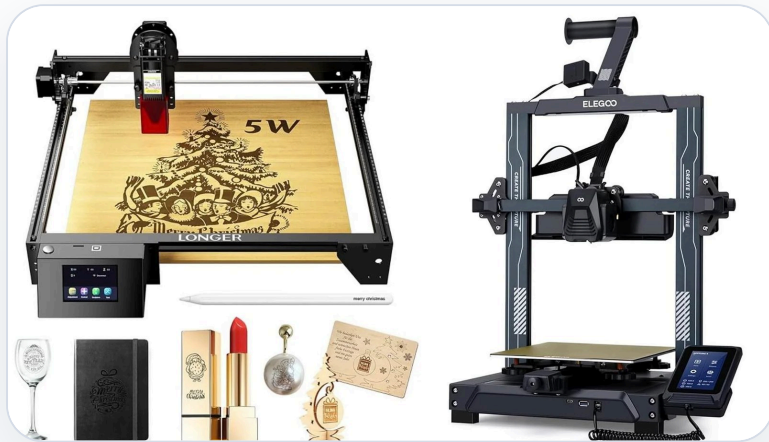


## MORE VIDEOS

A longer Onshape playlist is on the class page and in the syllabus.

# Two machines, two ways of making

Both are digital fabrication, but they remove material differently, so they suit different parts.

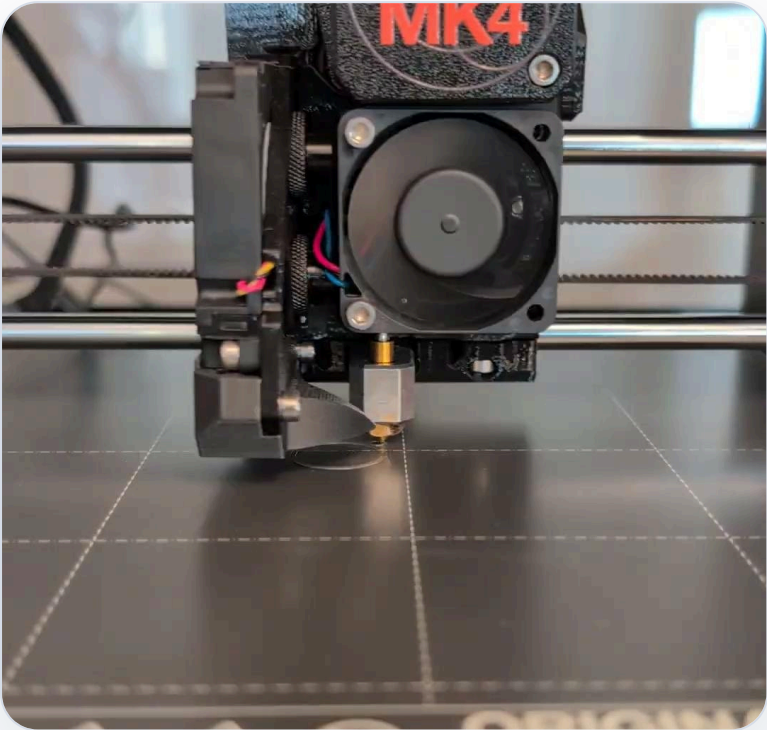


A laser cutter (left) and a 3D printer (right).

| Aspect     | 3D printing                                    | Laser cutting                                        |
|------------|------------------------------------------------|------------------------------------------------------|
| Process    | Additive. Builds the object layer by layer     | Subtractive. Removes material from a flat sheet      |
| Material   | Thermoplastics, resins, and metal powders      | Wood, acrylic, paper, fabric, and some plastics      |
| Complexity | Intricate, organic 3D geometry                 | Best for 2D. 3D shapes are assembled from flat parts |
| Precision  | Good resolution; layer lines affect the finish | High precision and tight tolerances on flat stock    |

# Choosing the right machine

Speed, cost, and finishing decide it, and often the answer is to use both.



The 3D printer builds up; the laser cuts down.

| Aspect    | 3D printing                                            | Laser cutting                                         |
|-----------|--------------------------------------------------------|-------------------------------------------------------|
| Speed     | Slower on large or highly detailed objects             | Fast from sheet stock. Excellent for batch runs       |
| Cost      | Cost-effective for one-off prototypes and custom parts | Lower cost per part from standard sheet sizes         |
| Finishing | Supports to remove, then sanding or finishing          | Minimal. Usually only cleaning the edges              |
| Best for  | Rapid prototypes and complex 3D models                 | Signage, panels, precision parts, high-volume cutting |

# Where the file meets the machine

The control software is both the design environment and the machine controller.

## Import

Vector and image files. SVG, DXF, AI, PDF, PNG, JPG, and more.

## Prepare

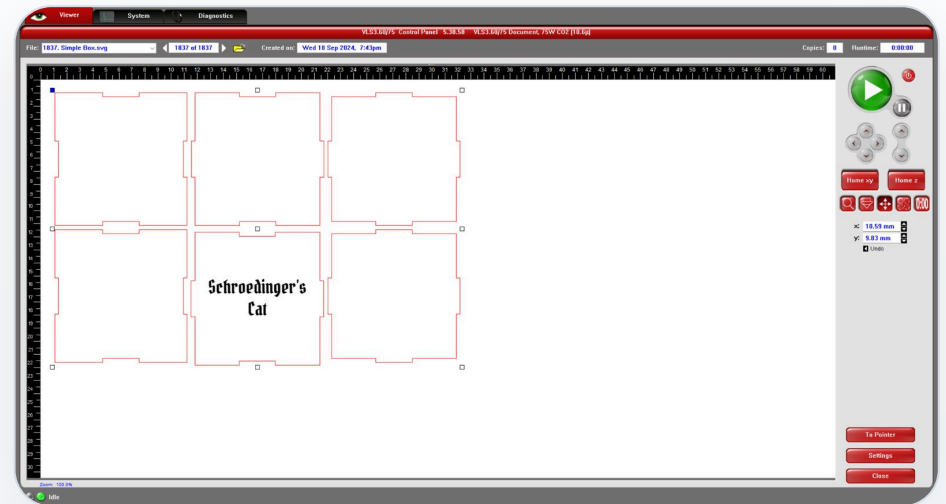
Scale, align, and add text so the job fits the sheet you actually have.

## Assign operations

Say which lines are cuts and which areas are engravings, with their own speeds and power levels.

## Send

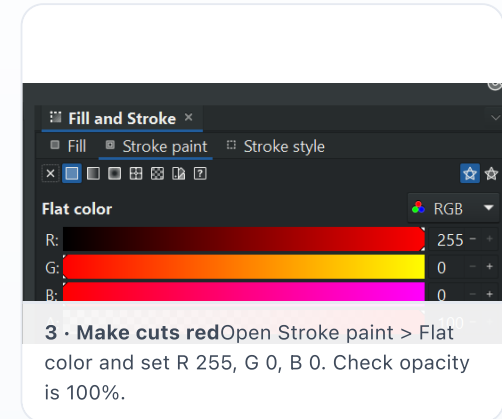
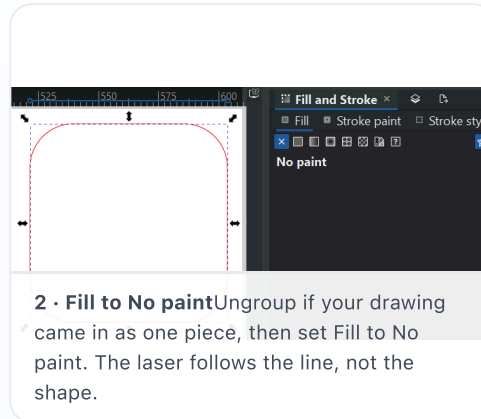
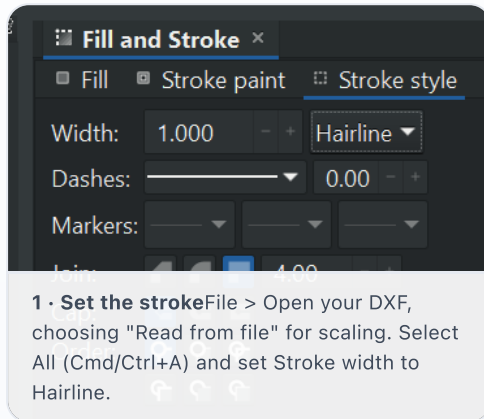
Where supported, it talks to the cutter directly. No separate driver step.



Red tool paths on the work plane, ready to send to the cutter.

# Open it in Inkscape and **fix the lines**

Your DXF provides the outline of what needs to be laser cut. The colors determine how you cut those lines. Red means cut all the way through. Blue means score.

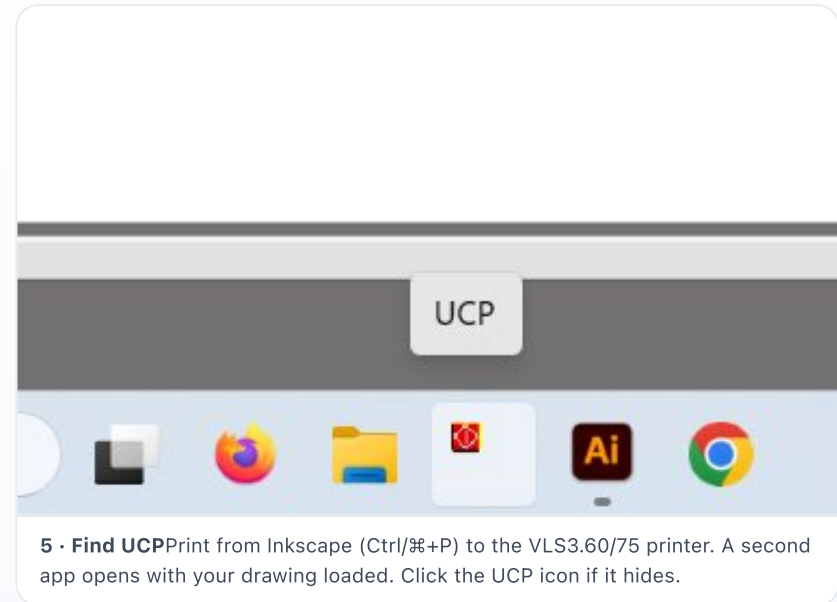
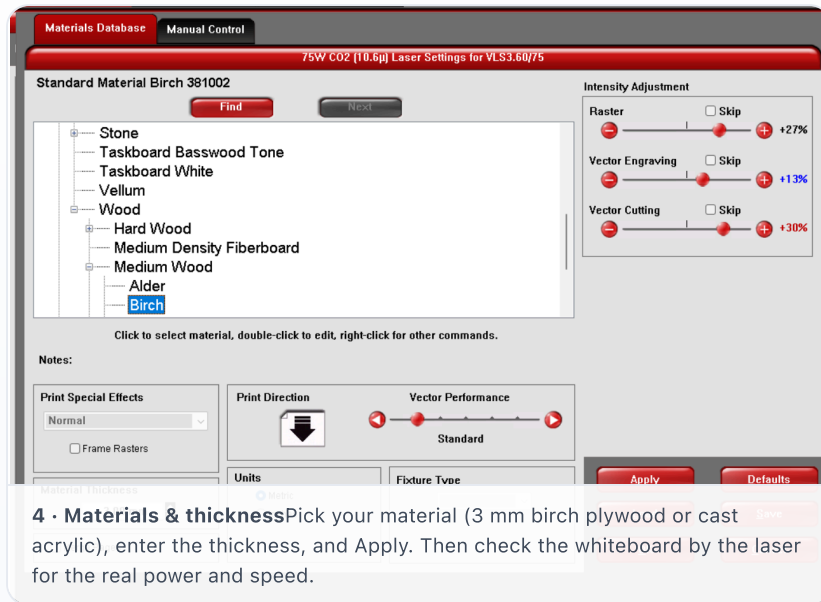


## THE RULE

**Cuts are 100% red. Scores are 100% blue. Engraves and text are 100% black.** The laser reads colour, so anything in the wrong colour is the wrong operation.

# Print it and hand it off

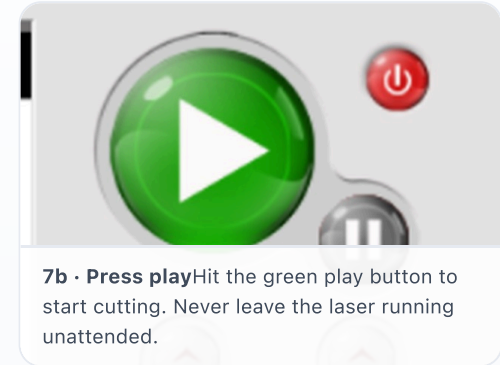
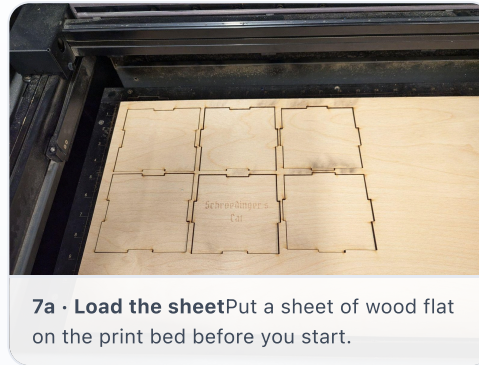
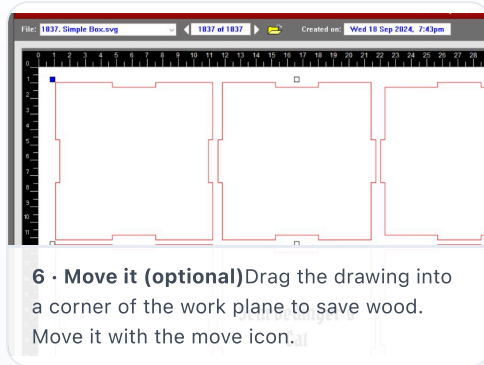
Inkscape is the editor; UCP runs the machine. Two steps to move between them.



**CHECK** Confirm your drawing arrived complete. Every cut line red, every score blue, every engrave black, nothing missing from the sheet.

# Place it, load it, **cut it**

Save material, load the sheet, and press go.



**SAFETY** Know where the **fire extinguisher** and the **emergency stop** are before you press play, and stay with the machine until the job finishes.

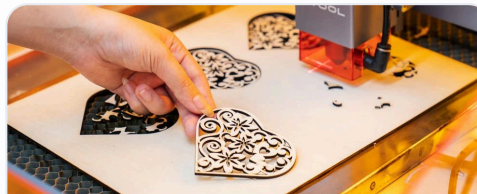
# What good looks like

Flat sheets becoming real, structural objects. The same jump your robot parts make.



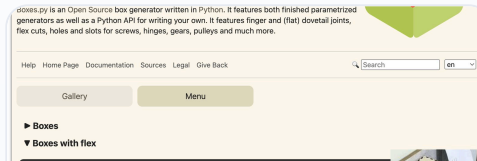
## Flat stock, 3D objects

Every one of these started as a 2D drawing.



## Fits matter

Cut a test part in cardboard first.



## Generate, then edit

Start from a template, finish it in Onshape.



## Cut and engrave

One job can cut lines and etch artwork.



## Sheets in, parts out

Plan the layout to use the whole sheet.

# Create an interesting **laser-cut object**

Anything you like. As long as you designed the vector file and cut it yourself.

01

## Design it

Draw your own vector file in Onshape, or start from a template you found, and remix it into something new.

02

## Cut it

Export the DXF, prepare it in Inkscape, and cut it at Nolop. Cut a cardboard test first.

03

## Assemble it

Build the object. Slots, glue, or bolts, washers and nuts.

### DELIVERABLES

**Submit your vector design files and upload photos of what you created.** Bring the assembled object to class for a demo.

### GUIDE

The full walkthrough. Install Inkscape, import your DXF, set the colours, run UCP. Is at:  
[tuftsmaker.github.io/ENT-164/laser-cutting/](https://tuftsmaker.github.io/ENT-164/laser-cutting/)

BEFORE YOU GO

# Questions?

Next week: From Physical to Digital: 3D. Model 3D parts in Onshape, prepare them in PrusaSlicer, and print the functional parts your robot needs.

## WHERE TO FIND US

### Office hours

[go.tufts.edu/meet\\_tvdv](https://go.tufts.edu/meet_tvdv)

### Email

[thomas.van\\_de\\_velde@tufts.edu](mailto:thomas.van_de_velde@tufts.edu)

### Course materials

[tuftsmaker.github.io/ENT-164](https://tuftsmaker.github.io/ENT-164)

# When you need a box, not a design

Generate the panels, then take them into Onshape or straight to the laser.

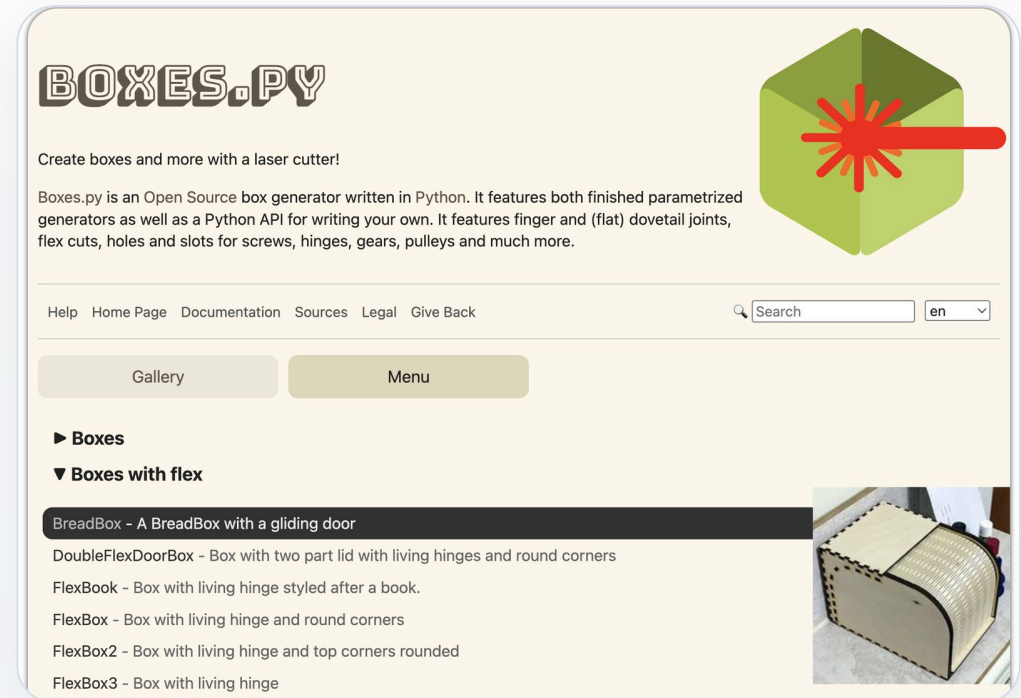
## LASER CUT SHAPE GENERATOR

Pick a box, a tray, or a living hinge, type your material thickness, and download an SVG.

[boxes.hackerspace-bamberg.de/boxes.py](https://boxes.hackerspace-bamberg.de/boxes.py)

## The material thickness drives everything

Every finger joint is sized from the sheet thickness. Measure it with your calipers first.



Boxes.py generates finger-jointed panels from your thickness.